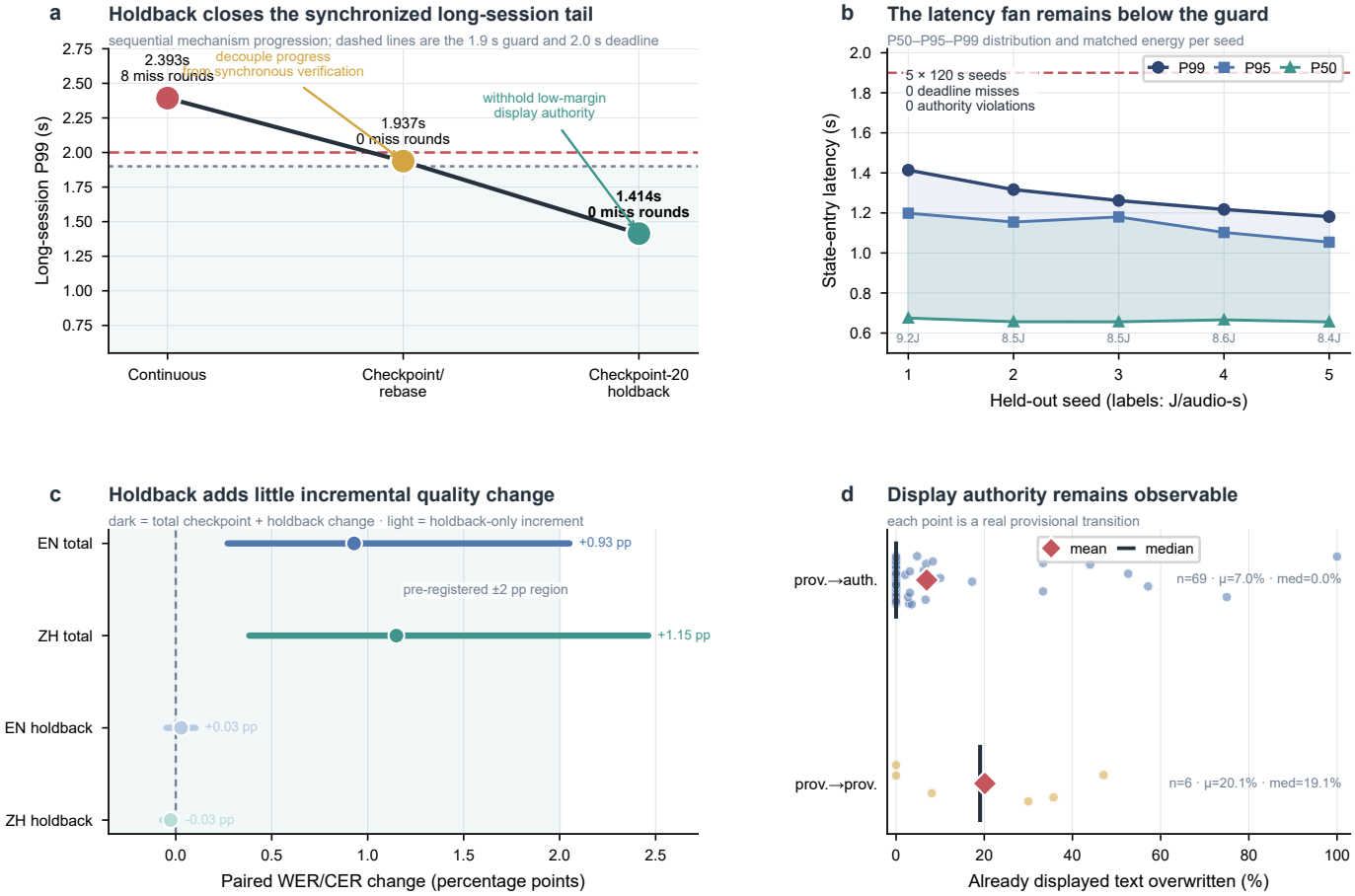

a
Holdback closes the synchronized long-session tail
sequential mechanism progression; dashed lines are the 1.9 s guard and 2.0 s deadline
Long-session P99 (s)
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
2.393s
8 miss rounds
decouple progress
from synchronous verification
1.937s
0 miss rounds
withhold low-margin
display authority
1.414s
0 miss rounds
Continuous
Checkpoint/
rebase
Checkpoint-20
holdback
b
The latency fan remains below the guard
P50–P95–P99 distribution and matched energy per seed
5 × 120 s seeds
0 deadline misses
0 authority violations
P99
P95
P50
State-entry latency (s)
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
9.2J
8.5J
8.5J
8.6J
8.4J
1
2
3
4
5
Held-out seed (labels: J/audio-s)
c
Holdback adds little incremental quality change
dark = total checkpoint + holdback change · light = holdback-only increment
EN total
ZH total
EN holdback
ZH holdback
+0.93 pp
pre-registered ±2 pp region
+1.15 pp
+0.03 pp
-0.03 pp
0.0
0.5
1.0
1.5
2.0
2.5
Paired WER/CER change (percentage points)
d
Display authority remains observable
each point is a real provisional transition
mean
median
prov.→auth.
n=69 · μ=7.0% · med=0.0%
prov.→prov.
n=6 · μ=20.1% · med=19.1%
0
20
40
60
80
100
Already displayed text overwritten (%)